

AI Model Routing Policy

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TL;DR — Route every AI workload to the cheapest approved model that meets its data classification and capability needs. Foundry is the default lane; routing is how cost, compliance and resilience become enforceable rather than advisory.

1. Purpose and problem statement

This policy defines how Mace Consult selects and routes AI model workloads. Version 2 adds the business case absent from v1: why routing exists, what it protects, and why Microsoft Foundry is the default routing lane. It responds to three problems that grow with adoption:

Cost. Current model prices span roughly 25x between tiers (Claude Haiku 4.5 at \$1/\$5 per million input/output tokens versus Claude Opus 4.8 at \$5/\$25; frontier tiers such as Claude Fable 5 list at \$10/\$50). Batch processing halves cost and prompt caching cuts repeated context by up to 90% — but only if workloads are deliberately routed. Billing is per invocation and per token: without routing, spend scales invisibly with success. The more AI is adopted, the worse un-routed spend becomes.

Data classification. Not every model deployment is approved for every data class. Routing turns “client-confidential data goes only to approved endpoints” from guidance into an enforceable control, aligned with the G2 security and DLP gate in the tiered environments model.

Resilience and lifecycle. Models are deprecated on short notice cycles, rate-limited, and occasionally unavailable. Routing, together with the model-pinning register, is what prevents a model retirement from breaking a production application — a failure mode Mace has already experienced in the form of blocked AI change requests.

2. Use cases

Use case	Workload shape	Routing outcome
MaceStyle document validation	High-volume, repeated rule context, non-interactive	Small model + prompt caching + batch; frontier models add cost, not accuracy
Project chatbots (MaceyBot/ZoeBot)	Conversational, M365-grounded, internal data	Copilot Studio lane; escalate complex tasks to governed agents
Document summarisation at scale	Bulk, asynchronous, mixed data classes	Routed by data class; batch pricing; caching on shared context
Complex analysis / agentic workflows	Low-volume, high-capability, multi-step	Frontier model justified; per-invocation cost logged and capped

3. Business case

3.1 Worked example — MaceStyle validation (annualised)

Assumptions: 2,000 validations per month; ~15K input tokens per document (of which ~10K is reusable rule context) and ~2K output tokens. Substitute actual volumes from production telemetry; the ratio, not the absolute figure, is the point. Prices are Anthropic list rates, July 2026, and apply equivalently via Foundry.

Routing approach	Cost per document	Annual cost
Un-routed: frontier model (Opus-class, \$5/\$25)	~\$0.125	~\$3,000
Routed: small model (Haiku-class, \$1/\$5)	~\$0.025	~\$600
Routed + rule context cached (90% off cache reads)	~\$0.016	~\$385

Routed + cached + batch (50% off, overnight runs)	~\$0.008	~\$190
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A single modest workload shows a ~15x spread between the default and the routed configuration. Applied across a portfolio of agents and per-invocation Copilot Credits, routing is the difference between AI spend that scales with value and spend that merely scales.

3.2 Summary of benefits

Dimension	Without routing	With routing
Cost	Every workload defaults to the model someone picked; spend grows with adoption	Cheapest sufficient model per task; caching and batch by design; per-app attribution and caps
Compliance	Data-class rules are advisory	Approved endpoint per data class, enforced at the routing layer and evidenced at G2
Resilience	Model retirement or outage breaks apps reactively	Pinning register, notice-period handling, fallback lanes tested in staging

4. The routing framework

Three tiers, unchanged from v1, route workloads by data classification and interaction pattern:

Tier	Route	Approved for	Notes
1	Azure OpenAI (GPT family)	Personal, client and regulated data	UK South / EU data zones; data residency enforced
2	Claude on Foundry (Zero Data Retention)	Internal, non-personal workloads	ZDR configuration; preferred for document intelligence and agentic tasks
3	Copilot Studio	Conversational assistants on M365 data	Permission-trimmed via Work IQ; maker-built agents subject to gate criteria

5. Why Foundry is the default lane

Mace operates a Microsoft estate: Entra ID identity, Azure billing, Purview compliance, and an existing enterprise agreement. Foundry provides access to multiple model families — including Claude and GPT models — under the commercial, security and data-residency terms Mace already holds: one procurement relationship rather than several, and one control plane shared with the Microsoft Agent Framework workloads on the platform roadmap.

Foundry is the **default** lane, not an exclusive one. Direct Anthropic API access remains for capabilities Foundry does not carry, with Bedrock and Vertex retained as resilience fallbacks. Vendor concentration is a managed risk, reviewed alongside the model exposure matrix.

6. Governance and review

Routing decisions are recorded per application in the model-pinning register. New workloads declare their data classification and expected volume at G1; routing configuration is evidenced at G2; cost-per-invocation telemetry is a release criterion at G3. This policy is owned by the Digital AI Service, reviewed quarterly or on material model, pricing or platform changes.

v2.0 responds to review feedback (A. — v1 lacked use case and business case). v1 framework content is retained unchanged in section 4. Prices sourced from published Anthropic rates, July 2026; verify at claude.com/pricing before external use.